Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-19 EM Wide Berm 4% - Test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.359 Hz*
Freq. upper bound:	*7.188 Hz*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*24*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*27*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *33.2%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1229*	*0.04085*
Peak wave period T_P (s):	*1.391*	*1.488*
Energy wave period $T_{-1,0}$ (s):	*1.34*	*1.208*
Mean wave period $T_{0,1}$ (s):	*1.256*	*0.9752*
Mean wave period $T_{0,2}$ (s):	*1.197*	*0.8553*
Spectral width (Broadness):	*0.7129*	*0.8079*
Spectral width (Narrowness):	*0.3189*	*0.5477*

Time Domain Analysis

Overall amplitude reflection coefficient: *33.3%*

	Incident	**Reflected**
Number of waves:	1001	1235
Mean wave height H_m (m):	*0.08123*	*0.02684*
Mean wave period T_m (s):	*1.278*	*1.036*
Sig. wave height H_s (m):	*0.1222*	*0.04063*
$T_{H_{1/3}}$ (s):	*1.379*	*1.256*
H_{max} (m):	*0.1909*	*0.0716*
T_{Hmax} (s):	*1.515*	*0.585*
$H_{1/10}$ (m):	*0.1461*	*0.04999*
$H_{1/50}$ (m):	*0.1671*	*0.05924*
$H_{1/100}$ (m)	*0.1742*	*0.06344*
$H_{1/250}$ (m):	*0.1837*	*0.06681*
$H_{0.1\%}$ (m):	*0.1909*	*0.07053*
$H_{1\%}$ (m):	*0.1655*	*0.05939*
$H_{2\%}$ (m):	*0.1558*	*0.05326*
$H_{10\%}$ (m)	*0.129*	*0.04301*
Groupiness factor GF:	*0.9199*	*1.029*
Skewness b1:	*0.4117*	*0.3509*
Kurtosis b2:	*2.82*	*3.122*

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Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-19 EM Wide Berm 4% - Test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.336 Hz*
Freq. upper bound:	*6.719 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*25*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*29*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *35%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1389*	*0.04855*
Peak wave period T_P (s):	*1.488*	*1.524*
Energy wave period $T_{-1,0}$ (s):	*1.405*	*1.284*
Mean wave period $T_{0,1}$ (s):	*1.3*	*1.003*
Mean wave period $T_{0,2}$ (s):	*1.225*	*0.8622*
Spectral width (Broadness):	*0.7629*	*0.8293*
Spectral width (Narrowness):	*0.3558*	*0.5935*

Time Domain Analysis

Overall amplitude reflection coefficient: *35%*

	Incident	Reflected
Number of waves:	993	1245
Mean wave height H_m (m):	*0.09174*	*0.03167*
Mean wave period T_m (s):	*1.345*	*1.073*
Sig. wave height H_s (m):	*0.1364*	*0.04812*
$T_{H_{1/3}}$ (s):	*1.465*	*1.356*
H_{max} (m):	*0.1951*	*0.0899*
T_{Hmax} (s):	*1.638*	*1.623*
$H_{1/10}$ (m):	*0.1595*	*0.05951*
$H_{1/50}$ (m):	*0.1803*	*0.07013*
$H_{1/100}$ (m)	*0.1857*	*0.07335*
$H_{1/250}$ (m):	*0.1902*	*0.07787*
$H_{0.1\%}$ (m):	*Not enough data*	*0.087*
$H_{1\%}$ (m):	*0.1814*	*0.06925*
$H_{2\%}$ (m):	*0.1684*	*0.06508*
$H_{10\%}$ (m)	*0.1445*	*0.05151*
Groupiness factor GF:	*0.8876*	*0.9916*
Skewness b1:	*0.485*	*0.2836*
Kurtosis b2:	*2.749*	*2.991*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-19 EM Wide Berm 4% - Test 1.experiment.dat*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.328 Hz*
Freq. upper bound:	*6.563 Hz*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*26*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*29*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *36.4%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1467*	*0.05336*
Peak wave period T_P (s):	*1.524*	*1.6*
Energy wave period $T_{-1,0}$ (s):	*1.446*	*1.338*
Mean wave period $T_{0,1}$ (s):	*1.33*	*1.026*
Mean wave period $T_{0,2}$ (s):	*1.246*	*0.874*
Spectral width (Broadness):	*0.7802*	*0.8341*
Spectral width (Narrowness):	*0.3733*	*0.6153*

Time Domain Analysis

Overall amplitude reflection coefficient: *36.4%*

	Incident	**Reflected**
Number of waves:	987	1231
Mean wave height H_m (m):	*0.09784*	*0.0353*
Mean wave period T_m (s):	*1.385*	*1.111*
Sig. wave height H_s (m):	*0.1449*	*0.05351*
$T_{H_{1/3}}$ (s):	*1.519*	*1.413*
H_{max} (m):	*0.2037*	*0.08502*
T_{Hmax} (s):	*1.587*	*0.8792*
$H_{1/10}$ (m):	*0.168*	*0.06517*
$H_{1/50}$ (m):	*0.1875*	*0.0754*
$H_{1/100}$ (m)	*0.1943*	*0.07806*
$H_{1/250}$ (m):	*0.2004*	*0.08178*
$H_{0.1\%}$ (m):	*Not enough data*	*0.0846*
$H_{1\%}$ (m):	*0.186*	*0.07466*
$H_{2\%}$ (m):	*0.1788*	*0.07156*
$H_{10\%}$ (m)	*0.1531*	*0.05666*
Groupiness factor GF:	*0.8769*	*0.981*
Skewness b1:	*0.5349*	*0.242*
Kurtosis b2:	*2.734*	*2.927*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-19 EM Wide Berm 4% - Test 1.experimentvelocity.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.320 Hz*
Freq. upper bound:	*6.406 Hz*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*27*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*30*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *37.8%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1575*	*0.05957*
Peak wave period T_P (s):	*1.561*	*1.73*
Energy wave period $T_{-1,0}$ (s):	*1.502*	*1.399*
Mean wave period $T_{0,1}$ (s):	*1.366*	*1.046*
Mean wave period $T_{0,2}$ (s):	*1.267*	*0.8783*
Spectral width (Broadness):	*0.8084*	*0.8445*
Spectral width (Narrowness):	*0.4024*	*0.6469*

Time Domain Analysis

Overall amplitude reflection coefficient: *37.8%*

	Incident	**Reflected**
Number of waves:	995	1272
Mean wave height H_m (m):	*0.1064*	*0.03936*
Mean wave period T_m (s):	*1.434*	*1.122*
Sig. wave height H_s (m):	*0.154*	*0.06017*
$T_{H_{1/3}}$ (s):	*1.588*	*1.475*
H_{max} (m):	*0.2118*	*0.1063*
T_{Hmax} (s):	*1.5*	*1.651*
$H_{1/10}$ (m):	*0.1767*	*0.07301*
$H_{1/50}$ (m):	*0.1972*	*0.08588*
$H_{1/100}$ (m)	*0.2019*	*0.09108*
$H_{1/250}$ (m):	*0.2077*	*0.09724*
$H_{0.1\%}$ (m):	*Not enough data*	*0.1043*
$H_{1\%}$ (m):	*0.1959*	*0.08424*
$H_{2\%}$ (m):	*0.1896*	*0.07894*
$H_{10\%}$ (m)	*0.1618*	*0.06413*
Groupiness factor GF:	*0.8367*	*0.9853*
Skewness b1:	*0.5571*	*0.2198*
Kurtosis b2:	*2.642*	*2.951*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-19 EM Wide Berm 4% - Test 1.experiment.dat*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.305 Hz*
Freq. upper bound:	*6.094 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*28*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*32*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *39.3%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1626*	*0.06381*
Peak wave period T_P (s):	*1.641*	*1.882*
Energy wave period $T_{-1,0}$ (s):	*1.546*	*1.487*
Mean wave period $T_{0,1}$ (s):	*1.398*	*1.105*
Mean wave period $T_{0,2}$ (s):	*1.292*	*0.9189*
Spectral width (Broadness):	*0.8135*	*0.8543*
Spectral width (Narrowness):	*0.4126*	*0.6688*

Time Domain Analysis

Overall amplitude reflection coefficient: *39.2%*

	Incident	Reflected
Number of waves:	1006	1227
Mean wave height H_m (m):	*0.1101*	*0.0423*
Mean wave period T_m (s):	*1.478*	*1.212*
Sig. wave height H_s (m):	*0.1579*	*0.06448*
$T_{H_{1/3}}$ (s):	*1.645*	*1.589*
H_{max} (m):	*0.2217*	*0.103*
T_{Hmax} (s):	*1.475*	*1.984*
$H_{1/10}$ (m):	*0.1813*	*0.07721*
$H_{1/50}$ (m):	*0.2012*	*0.0883*
$H_{1/100}$ (m)	*0.2075*	*0.09171*
$H_{1/250}$ (m):	*0.2144*	*0.09593*
$H_{0.1\%}$ (m):	*0.2216*	*0.1017*
$H_{1\%}$ (m):	*0.1997*	*0.08669*
$H_{2\%}$ (m):	*0.1932*	*0.08286*
$H_{10\%}$ (m)	*0.1654*	*0.06908*
Groupiness factor GF:	*0.8345*	*0.9694*
Skewness b1:	*0.5821*	*0.2159*
Kurtosis b2:	*2.633*	*2.851*

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Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-19 EM Wide Berm 4% - Test 1.experimentvelfjord*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.266 Hz*
Freq. upper bound:	*5.313 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*28*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*36*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *40.2%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1644*	*0.06608*
Peak wave period T_P (s):	*1.778*	*1.882*
Energy wave period $T_{-1,0}$ (s):	*1.587*	*1.567*
Mean wave period $T_{0,1}$ (s):	*1.423*	*1.156*
Mean wave period $T_{0,2}$ (s):	*1.311*	*0.9551*
Spectral width (Broadness):	*0.8032*	*0.8499*
Spectral width (Narrowness):	*0.4218*	*0.6812*

Time Domain Analysis

Overall amplitude reflection coefficient: *40.2%*

	Incident	Reflected
Number of waves:	984	1103
Mean wave height H_m (m):	*0.1114*	*0.04521*
Mean wave period T_m (s):	*1.512*	*1.349*
Sig. wave height H_s (m):	*0.1607*	*0.06788*
$T_{H_{1/3}}$ (s):	*1.683*	*1.719*
H_{max} (m):	*0.2308*	*0.1141*
T_{Hmax} (s):	*1.769*	*1.274*
$H_{1/10}$ (m):	*0.1848*	*0.08047*
$H_{1/50}$ (m):	*0.2067*	*0.09223*
$H_{1/100}$ (m)	*0.2138*	*0.09815*
$H_{1/250}$ (m):	*0.2254*	*0.1048*
$H_{0.1\%}$ (m):	*Not enough data*	*0.1133*
$H_{1\%}$ (m):	*0.2017*	*0.08955*
$H_{2\%}$ (m):	*0.1976*	*0.0842*
$H_{10\%}$ (m)	*0.1689*	*0.07238*
Groupiness factor GF:	*0.8492*	*0.976*
Skewness b1:	*0.6059*	*0.1965*
Kurtosis b2:	*2.679*	*2.816*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-19 EM Wide Berm 4% - Test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.461 Hz*
Freq. upper bound:	*9.219 Hz*

Freq. Domain Analysis

Overlap (FFT) timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*19*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*21*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *28.1%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.0735*	*0.02068*
Peak wave period T_P (s):	*1.085*	*1.085*
Energy wave period $T_{-1,0}$ (s):	*1.06*	*0.9757*
Mean wave period $T_{0,1}$ (s):	*1.027*	*0.8542*
Mean wave period $T_{0,2}$ (s):	*1.006*	*0.7819*
Spectral width (Broadness):	*0.509*	*0.7452*
Spectral width (Narrowness):	*0.2059*	*0.4399*

Time Domain Analysis

Overall amplitude reflection coefficient: *28.1%*

	Incident	**Reflected**
Number of waves:	980	1149
Mean wave height H_m (m):	*0.0465*	*0.01334*
Mean wave period T_m (s):	*1.028*	*0.8763*
Sig. wave height H_s (m):	*0.07246*	*0.02011*
$T_{H_{1/3}}$ (s):	*1.052*	*1.013*
H_{max} (m):	*0.1315*	*0.0316*
T_{Hmax} (s):	*1.063*	*1.097*
$H_{1/10}$ (m):	*0.09173*	*0.02449*
$H_{1/50}$ (m):	*0.1098*	*0.02831*
$H_{1/100}$ (m)	*0.1167*	*0.02926*
$H_{1/250}$ (m):	*0.1242*	*0.03048*
$H_{0.1\%}$ (m):	*Not enough data*	*0.0315*
$H_{1\%}$ (m):	*0.1086*	*0.02774*
$H_{2\%}$ (m):	*0.1006*	*0.02662*
$H_{10\%}$ (m)	*0.07744*	*0.0214*
Groupiness factor GF:	*0.9739*	*0.9435*
Skewness b1:	*0.05002*	*0.08323*
Kurtosis b2:	*2.971*	*2.808*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-19 EM Wide Berm 4% - Test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.391 Hz*
Freq. upper bound:	*7.813 Hz*

Freq. Domain Analysis

Overlap (FFT):	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*22*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*25*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *31.9%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1056*	*0.03363*
Peak wave period T_P (s):	*1.28*	*1.28*
Energy wave period $T_{-1,0}$ (s):	*1.217*	*1.062*
Mean wave period $T_{0,1}$ (s):	*1.161*	*0.881*
Mean wave period $T_{0,2}$ (s):	*1.12*	*0.787*
Spectral width (Broadness):	*0.6766*	*0.7896*
Spectral width (Narrowness):	*0.2729*	*0.5034*

Time Domain Analysis

Overall amplitude reflection coefficient: *31.8%*

	Incident	**Reflected**
Number of waves:	991	1222
Mean wave height H_m (m):	*0.06828*	*0.02201*
Mean wave period T_m (s):	*1.162*	*0.942*
Sig. wave height H_s (m):	*0.1044*	*0.03341*
$T_{H_{1/3}}$ (s):	*1.234*	*1.11*
H_{max} (m):	*0.1715*	*0.06029*
T_{Hmax} (s):	*1.21*	*1.656*
$H_{1/10}$ (m):	*0.1287*	*0.04198*
$H_{1/50}$ (m):	*0.1509*	*0.05032*
$H_{1/100}$ (m)	*0.1577*	*0.05341*
$H_{1/250}$ (m):	*0.1659*	*0.05681*
$H_{0.1\%}$ (m):	*Not enough data*	*0.05954*
$H_{1\%}$ (m):	*0.1503*	*0.04929*
$H_{2\%}$ (m):	*0.1381*	*0.04621*
$H_{10\%}$ (m)	*0.1122*	*0.03535*
Groupiness factor GF:	*0.9275*	*1.08*
Skewness b1:	*0.1726*	*0.2768*
Kurtosis b2:	*2.877*	*3.235*